



Microbial community / environmental DNA analysis with Chipster

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What will I learn?

- Microbial community / environmental DNA analysis of amplicon sequencing data
 - Central concepts
 - Analysis steps
 - File formats
- Microbial 16S rRNA gene data is used in the exercises
 - We discuss also how to analyze data on other marker genes
- How to operate the Chipster software

Understanding data analysis - why?

- Bioinformaticians might not always be available when needed
- Biologists know their own experiments best
 - Potential batch effects etc
- Allows you to design experiments better → less money wasted
- Allows you to discuss more easily with bioinformaticians

Introduction to microbial community / eDNA analysis

Outline

- What questions does it answer
- How is it done
- What are the main steps

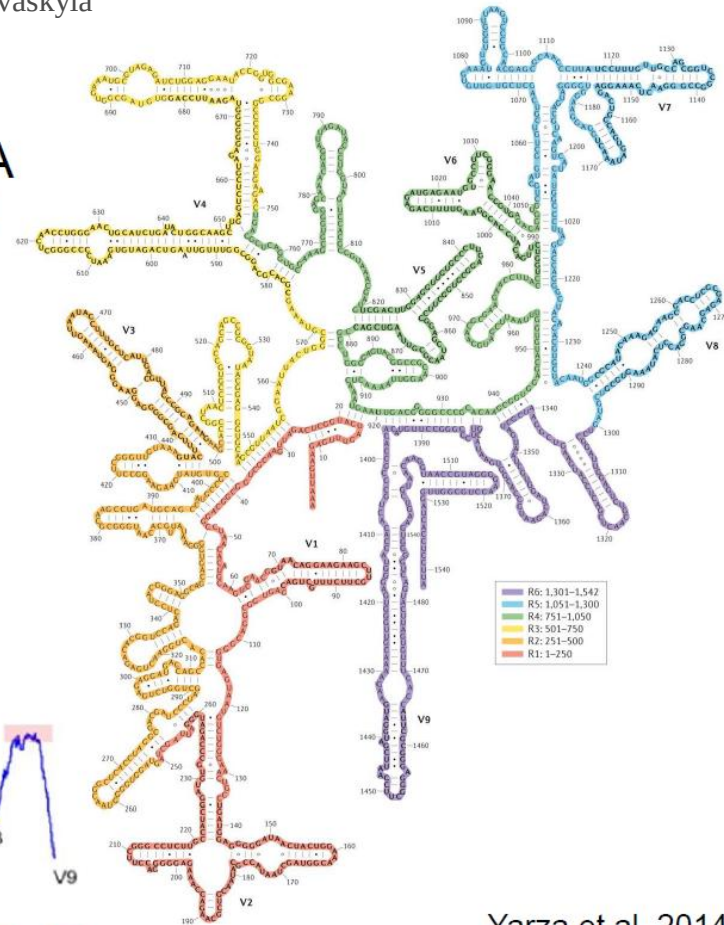
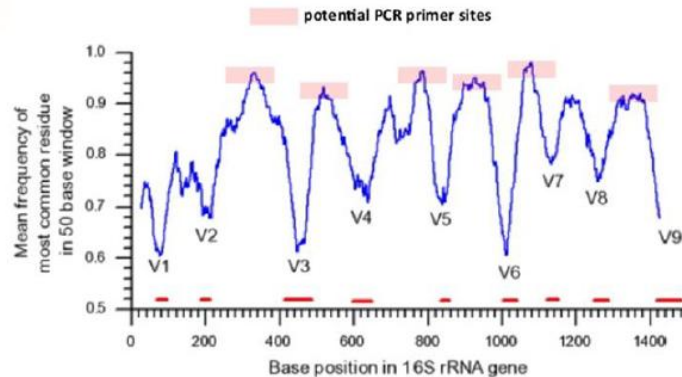
Microbial community / eDNA analysis

- Answers the questions **who are there** and **in what proportions** if compared to your other samples
 - It will not confirm that someone isn't there (sampling depth, primer/sequencing bias)
- Specific primers are used to amplify a region of one gene
 - Bacterial and archaeal communities: 16S rRNA
 - Fungal communities: ITS (internal transcribed spacer between 18S and 5.8S rRNA genes)
- Sequenced using Illumina MiSeq (or Ion Torrent)
 - Also long read sequencing with better resolution: PacBio, Oxford Nanopore
- Different from metagenomics, where the aim is to sequence all genes
 - Answers the questions **who are there** and **what are they capable of doing**

Commonly used 16S rRNA gene amplicons are called by the variable regions they contain

- V1-V2
- (V3)-V4

Variable Regions of the 16S rRNA:



Yarza et al. 2014

Nature Reviews | Microbiology

UNIVERSITY OF JYVÄSKYLÄ

Main parts of microbial community analysis (DADA2)



- **Preprocessing**

- Quality control, trim primers/adaptors and bad quality ends
- Filter and trim sequences based on quality scores and/or length

- **Processing reads into ASVs**

- Sample inference: separating sequencing errors from biological variation → ASVs
- Create contigs
- Remove chimeras
- Taxonomic assignment of ASVs (e.g. SILVA for 16S, UNITE for ITS)

- **Community analysis and visualization**

- Does community structure differ between sample groups?
- Which taxa are differentially abundant between sample groups?

How to choose the preprocessing protocol?

- Sequencing technology:
 - Illumina Miseq: paired-end short reads
 - Ion Torrent: single-end short reads
 - PacBio and Nanopore: long reads
- Gene: 16S rRNA, ITS, other?
 - reference database
 - gene characteristics
- Operational taxonomic units (OTUs) vs. amplicon sequence variants (ASVs)
 - OTUs: mothur, QIIME2
 - ASVs: DADA2

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 - **ASVs: DADA2**

OTUs vs. ASVs

- DADA2 pipeline and ASV-based analyses hugely popular
- ASV = amplicon sequence variant
 - each sequence is a different ASV
 - different copies of 16S rRNA gene from the same organism are different ASVs
- OTU = operational taxonomic unit
 - sequences clustered into 'species' based on a similarity cut off
 - bacterial 16S rRNA gene → 97 %
 - very closely related species based on short reads cluster into the same OTU

→ clump or split?

Data set for exercises: mouse fecal samples



- Subset of 19 mouse fecal samples
- Two time points:
 - early: collected on days 0-9 post weaning
 - late: collected on days 141-150 post weaning

→ Does mouse gut microbiome change?

Data set for exercises: mouse fecal samples



- PCR amplification of V₄ region of the bacterial 16S rRNA gene (ca. 250 bp)
- Sequencing by Illumina MiSeq
- Demultiplexed (separated by sample) and barcodes removed
- Primers and sequencing adapters removed
- Each sample in two separate FASTQ files:
 - forward reads (R1)
 - reverse reads (R2)

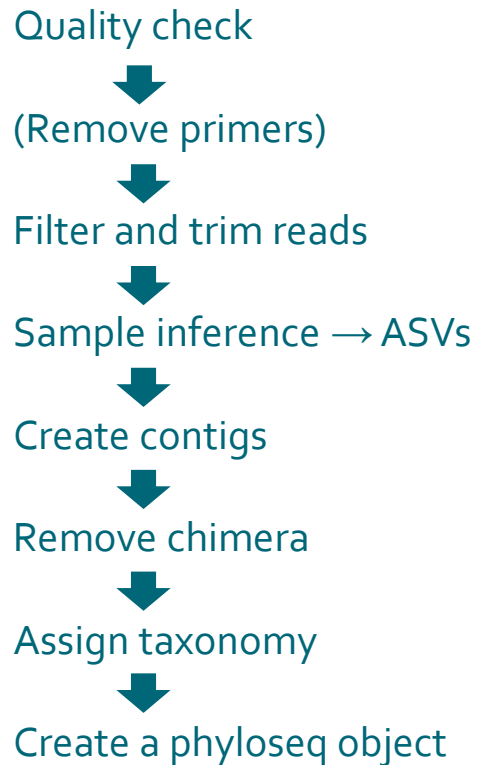
Processing data of other genes than 16S, 18S or ITS

- Similar pre-processing workflows can be followed as with 16S, 18S or ITS data, but you have to provide the following files:
 - OTU workflow: Mothur-formatted reference sequence FASTA file and taxonomy file
 - ASV workflow: DADA2-formatted reference sequence FASTA file
- Depending on the gene, you might have to skip the alignment step or align the sequences outside Chipster.

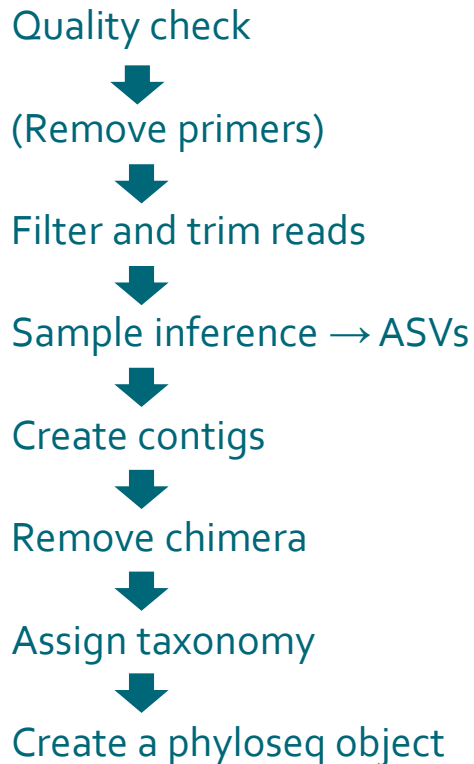
ASV-based analysis and DADA2 in Chipster

- Pre-processing: Microbial amplicon data preprocessing for ASV
 - Remove primers and adapters
 - Filter and trim
 - Sample (ASV) inference
 - Combine paired reads to contigs
 - Remove chimeras
 - Assign taxonomy
- ➡ Check videos on the Chipster Youtube channel for details
- ➡ All DADA2 materials:
<https://chipster.2.rahtiapp.fi/manual/courses.html#microbiome>

Outline of Day 1:



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Output so far: phyloseq object with

1. ASV table (counts in each sample)
2. taxonomy of ASVs
3. sample information (phenodata)
4. sequences of ASVs

Day 2:

- Data tidying & transformations
- Taxonomy plots
- Alpha diversity
- Beta diversity: ordinations & statistics